Jacobian Descent For Multi-Objective Optimization

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Joint work with Pierre Quinton

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GitHub
github.com/TorchJD/torchjd

TorchJD
torchjd.org

arXiv
arxiv.org/pdf/2406.16232

Table of contents

Theory
- Partial order
- Pareto optimality
- Jacobian descent
- Unconflicting projection of gradients (UPGrad)

Applications and experimentation
- Optimization trajectories
- Per-sample JD
- Multi-task learning

Conclusion

Theory

Partial order

Define the partial order $(\mathbb{R}^m, \preceq)$, for $\boldsymbol{u}, \boldsymbol{v} \in \mathbb{R}^m$, as

$$\begin{aligned} & \boldsymbol{u} \preceq \boldsymbol{v} & \\ \Leftrightarrow \qquad & u_i \leq v_i & \forall i \in [m] \end{aligned}$$

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Examples:

- $\begin{bmatrix} 2 \\ 1 \end{bmatrix} \preceq \begin{bmatrix} 2 \\ 2 \end{bmatrix}$

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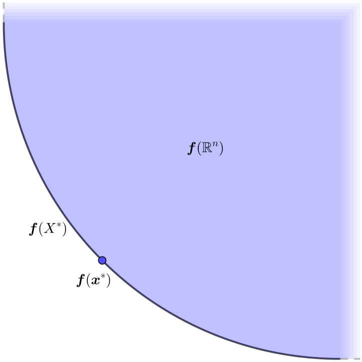
Examples:

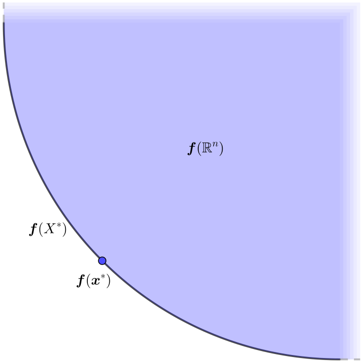
- $\begin{bmatrix} 2 \\ 1 \end{bmatrix} \preceq \begin{bmatrix} 2 \\ 2 \end{bmatrix}$
- $\begin{bmatrix} 2 \\ 1 \end{bmatrix}$ and $\begin{bmatrix} 1 \\ 2 \end{bmatrix}$ are not comparable with this partial order.

Pareto optimality



Pareto optimality

Pareto set: X^*



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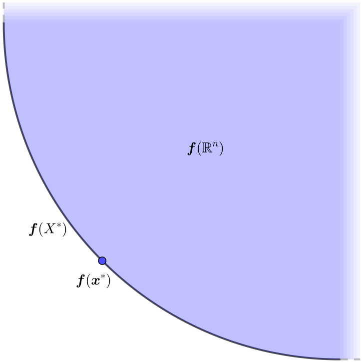

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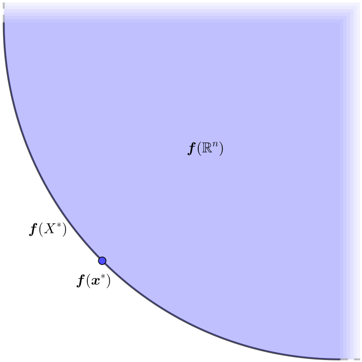

$$\boldsymbol{x}^* \in X^* \quad \Leftrightarrow \quad \forall \boldsymbol{y} \in \mathbb{R}^n,\ \boldsymbol{f}(\boldsymbol{y}) \preceq \boldsymbol{f}(\boldsymbol{x}^*) \rightarrow \boldsymbol{f}(\boldsymbol{y}) = \boldsymbol{f}(\boldsymbol{x}^*)$$

Jacobian descent

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$$\mathcal{J}\boldsymbol{f}(\boldsymbol{x}) = \begin{bmatrix} \nabla f_1(\boldsymbol{x})^\top \\ \nabla f_2(\boldsymbol{x})^\top \\ \vdots \\ \nabla f_m(\boldsymbol{x})^\top \end{bmatrix} = \begin{bmatrix} \frac{\partial}{\partial x_1} f_1(\boldsymbol{x}) & \frac{\partial}{\partial x_2} f_1(\boldsymbol{x}) & \cdots & \frac{\partial}{\partial x_n} f_1(\boldsymbol{x}) \\ \frac{\partial}{\partial x_1} f_2(\boldsymbol{x}) & \frac{\partial}{\partial x_2} f_2(\boldsymbol{x}) & \cdots & \frac{\partial}{\partial x_n} f_2(\boldsymbol{x}) \\ \vdots & \vdots & \ddots & \vdots \\ \frac{\partial}{\partial x_1} f_m(\boldsymbol{x}) & \frac{\partial}{\partial x_2} f_m(\boldsymbol{x}) & \cdots & \frac{\partial}{\partial x_n} f_m(\boldsymbol{x}) \end{bmatrix}$$

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For a small enough update $\boldsymbol{y} \in \mathbb{R}^n$, Taylor's theorem states

$$\boldsymbol{f}(\boldsymbol{x} + \boldsymbol{y}) \approx \boldsymbol{f}(\boldsymbol{x}) + \mathcal{J}\boldsymbol{f}(\boldsymbol{x}) \cdot \boldsymbol{y}$$

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$$\boldsymbol{f}(\boldsymbol{x} + \boldsymbol{y}) \approx \boldsymbol{f}(\boldsymbol{x}) + \mathcal{J}\boldsymbol{f}(\boldsymbol{x}) \cdot \boldsymbol{y}$$

We want $\boldsymbol{f}(\boldsymbol{x} + \boldsymbol{y}) \preceq \boldsymbol{f}(\boldsymbol{x})$. We select $\boldsymbol{y} = -\eta \mathcal{A}(\mathcal{J}\boldsymbol{f}(\boldsymbol{x})) \in \mathbb{R}^n$.

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The mapping $\mathcal{A} : \mathbb{R}^{m \times n} \to \mathbb{R}^n$ is called an **aggregator**, it parameterizes **Jacobian descent**:

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Algorithm 1: Jacobian descent with aggregator $\mathcal{A}$

Input: $\boldsymbol{x} \in \mathbb{R}^n$, $0 < \eta$, $T \in \mathbb{N}$, $\mathcal{A} : \mathbb{R}^{m \times n} \to \mathbb{R}^n$

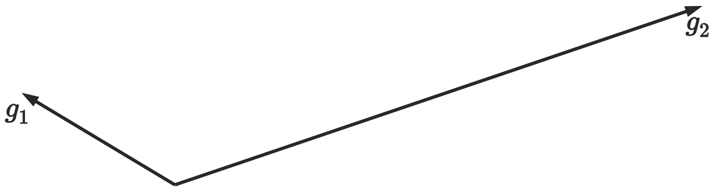
for $t \leftarrow 1$ **to** T **do**

$\quad \boldsymbol{x} \leftarrow \boldsymbol{x} - \eta \mathcal{A}(\mathcal{J}\boldsymbol{f}(\boldsymbol{x}))$

Output: $\boldsymbol{x}$

Unconflicting projection of gradients (UPGrad)

$$J = \begin{bmatrix} g_1^\top \\ g_2^\top \end{bmatrix}$$



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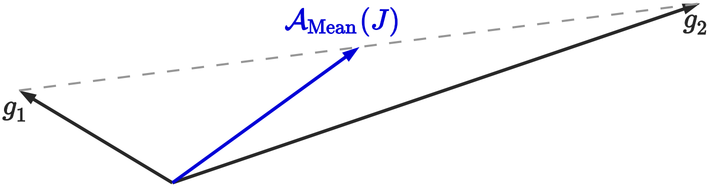

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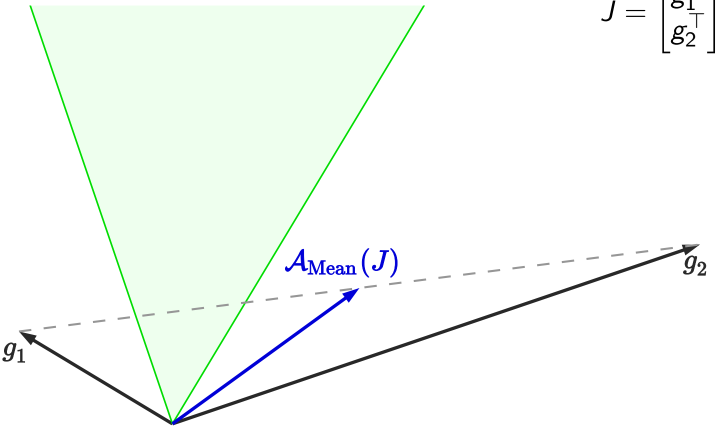

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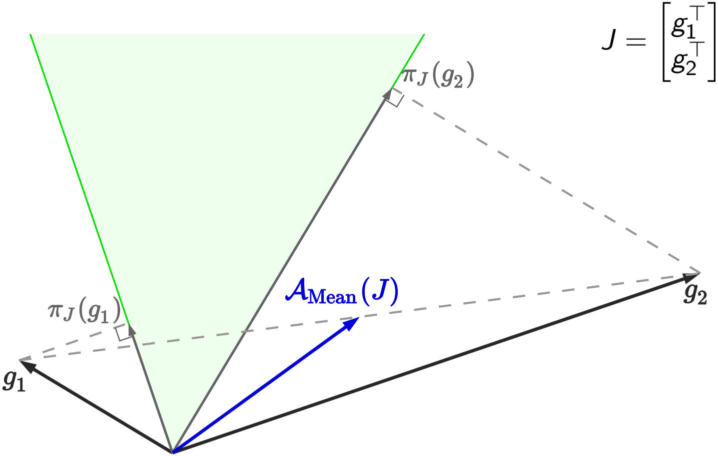

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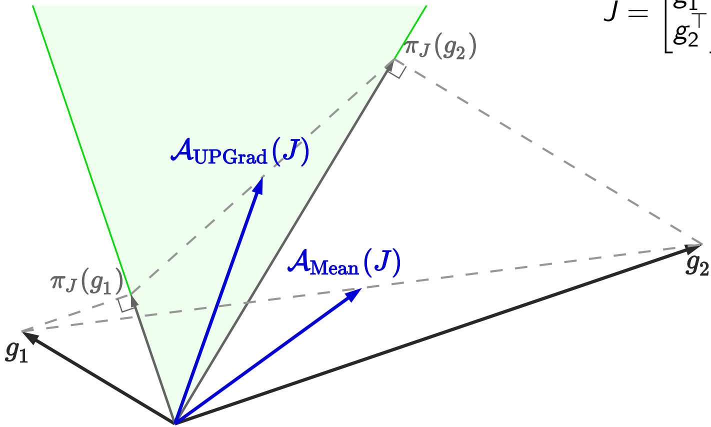
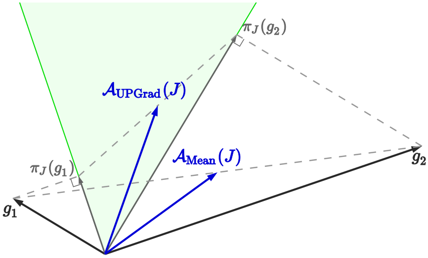


Formal definition of UPGrad

Given $J \in \mathbb{R}^{m \times n}$, let C_J be the cone $\{\boldsymbol{y} \in \mathbb{R}^n : \mathbf{0} \preceq J\boldsymbol{y}\}$.

$$\pi_J(\boldsymbol{x}) = \underset{\boldsymbol{y} \in C_J}{\arg\min} \|\boldsymbol{x} - \boldsymbol{y}\|^2$$

$$\mathcal{A}_{\text{UPGrad}}(J) = \frac{1}{m} \sum_{i \in [m]} \pi_J(g_i)$$



Applications and experimentation

Optimization trajectories

Goal: Minimize some convex quadratic function $\boldsymbol{f} : \mathbb{R}^2 \to \mathbb{R}^2$, with conflicting gradients.

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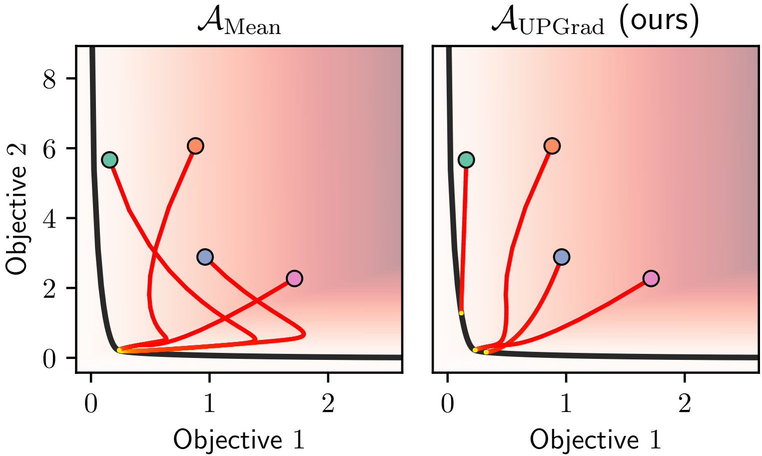

Optimization trajectories: code

```python
from torchjd import autojac
from torchjd.aggregation import UPGrad

x = tensor([0., 0.])  # initial value of x

for i in range(n_iters):
  y = f(x)  # shape: [2]
  autojac.backward(y)
  # x now has a .jac field
  autojac.jac_to_grad([x], UPGrad())
  # x now has a .grad field
  optimizer.step()
  optimizer.zero_grad()
```

Full project available at github.com/TorchJD/trajectories

Per-sample JD

Each element of the batch has its own loss.

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```
from torchjd import autojac
from torchjd.aggregation import UPGrad

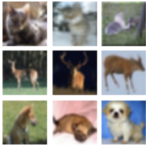
batch_size = 16
model = Linear(10, 1)

for x, y in dataloader:
  z = model(x).squeeze(dim=1)  # shape: [16]
  losses = loss_fn(z, y)  # shape: [16]
  autojac.backward(losses)
  # parameters now have a .jac field
  autojac.jac_to_grad(model.parameters(), UPGrad())
  # parameters now have a .grad field
  optimizer.step()
  optimizer.zero_grad()
```

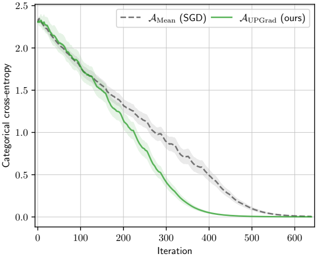
Full code example available at torchjd.org/latest/examples/iwrm.

Per-sample JD: results

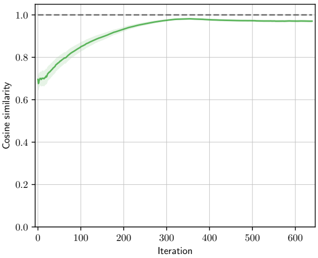
CIFAR-10



Training Loss



Update Similarity



Per-sample JD: results

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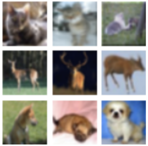
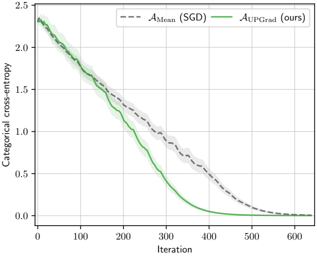
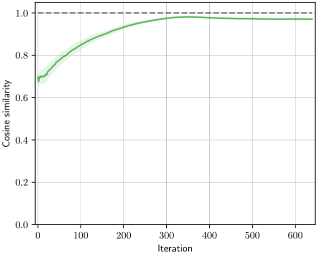
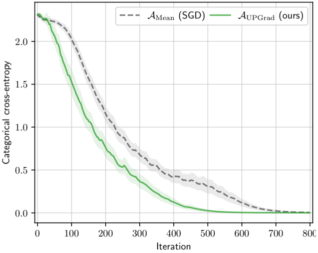
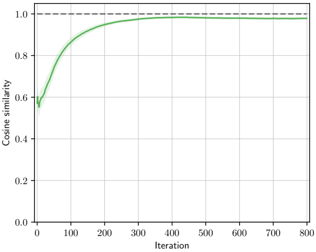






SVHN





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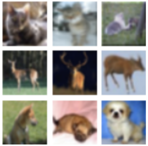

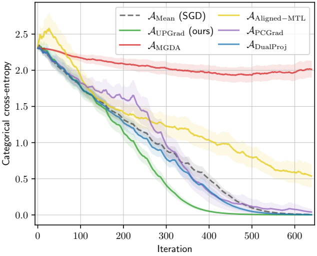

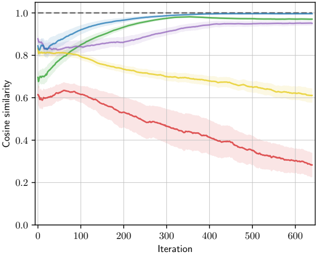

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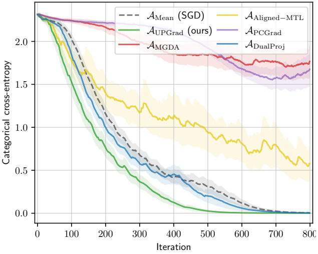

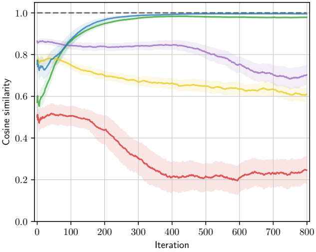

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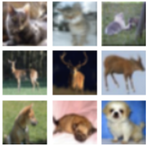

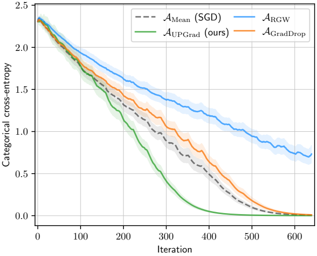

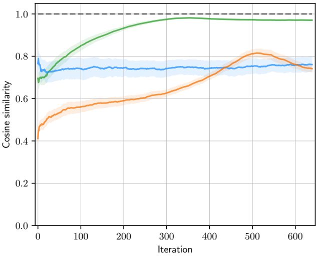

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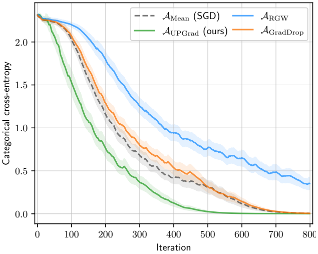

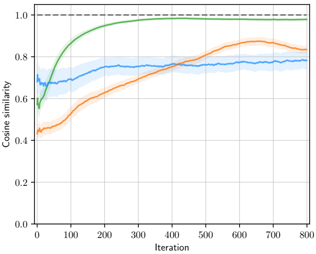

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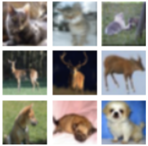

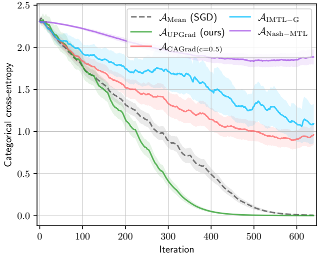

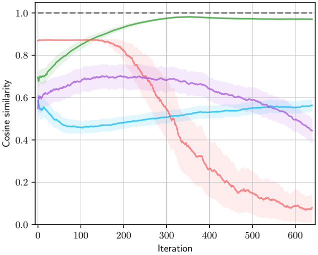

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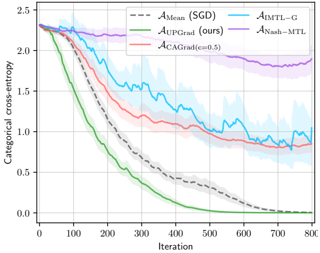

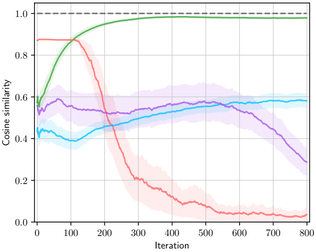

Multi-task learning

$$
\begin{tikzcd}
 & & & & & \boldsymbol{y}_1 \arrow[d] \\
 & & & \boxed{\boldsymbol{h}_1(\boldsymbol{r}, \boldsymbol{\theta}_1)} \arrow[r] & \boldsymbol{z}_1 \arrow[r] & \boxed{\ell_1(\boldsymbol{z}_1, \boldsymbol{y}_1)} \\
\boldsymbol{x} \arrow[r] & \boxed{\boldsymbol{f}(\boldsymbol{x}, \boldsymbol{\theta}_0)} \arrow[r] & \boldsymbol{r} \arrow[ur] \arrow[dr] & \vdots & \vdots & \vdots \\
 & & & \boxed{\boldsymbol{h}_m(\boldsymbol{r}, \boldsymbol{\theta}_m)} \arrow[r] & \boldsymbol{z}_m \arrow[r] & \boxed{\ell_m(\boldsymbol{z}_m, \boldsymbol{y}_m)} \\
 & & & & & \boldsymbol{y}_m \arrow[u]
\end{tikzcd}
$$

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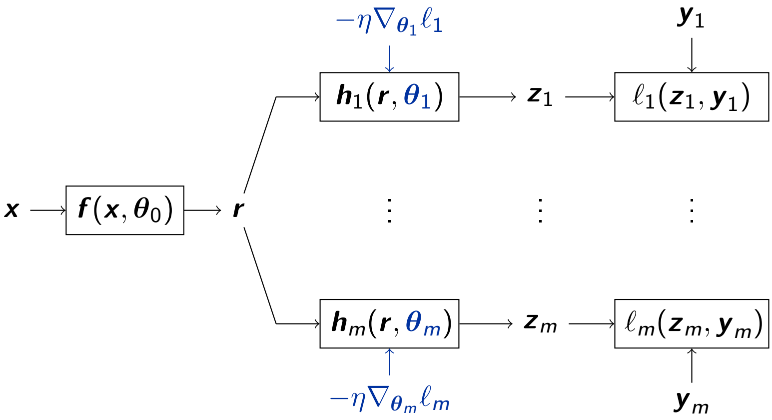

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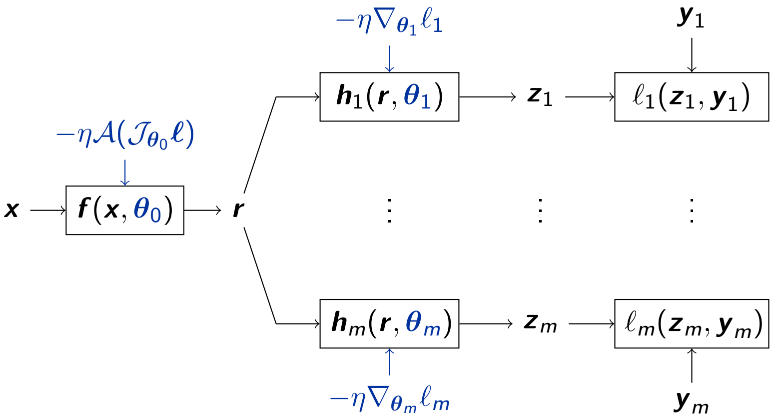

Multi-task learning: code

```
from torchjd import autojac
from torchjd.aggregation import UPGrad

f = Linear(10, 5)
h1 = Linear(5, 1)
h2 = Linear(5, 1)
for x, y1, y2 in dataloader:
  r = f(x)
  z1, z2 = h1(r), h2(r)
  l1, l2 = loss_fn(z1, y1), loss_fn(z2, y2)
  autojac.mtl_backward([l1, l2], features=r)
  # head parameters now have a .grad field
  # backbone parameters now have a .jac field
  autojac.jac_to_grad(f.parameters(), UPGrad())
  # All parameters now have a .grad field
  optimizer.step()
  optimizer.zero_grad()
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Full code example available at torchjd.org/latest/examples/mtl.

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Can JD solve problems that GD can't?

Appendix

Gramian-based JD

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The weights only depend on the (small) Gramian $\mathcal{G}\boldsymbol{f}(\boldsymbol{x}) = \mathcal{J}\boldsymbol{f}(\boldsymbol{x}) \cdot \mathcal{J}\boldsymbol{f}(\boldsymbol{x})^\top$.

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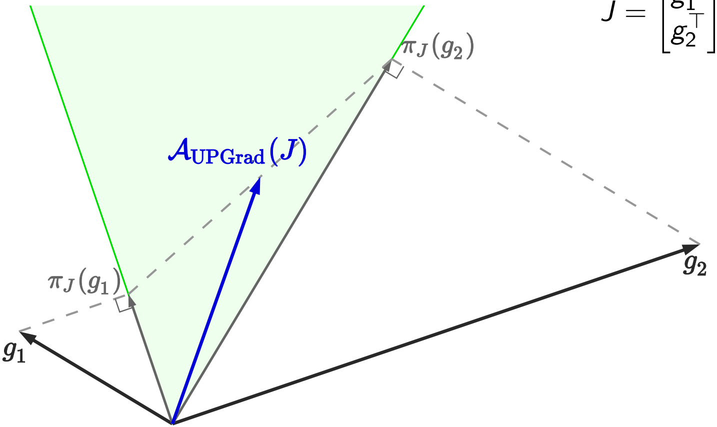
Applicable to most other aggregators from the literature.

Other aggregators

$$J = \begin{bmatrix} g_1^\top \\ g_2^\top \end{bmatrix}$$



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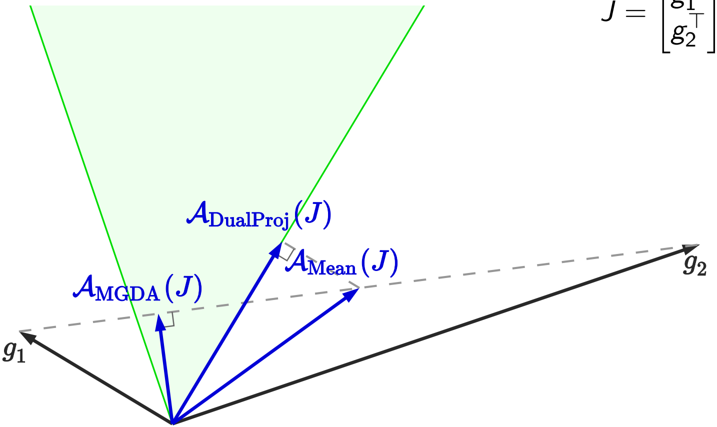

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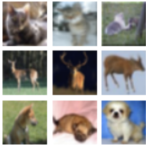
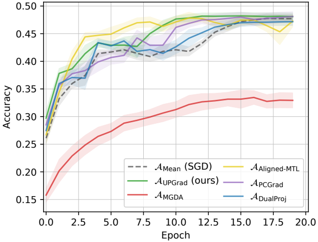
Per-sample JD: test accuracies

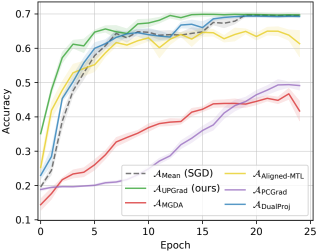
Test Accuracy

CIFAR-10





SVHN



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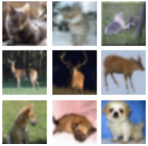

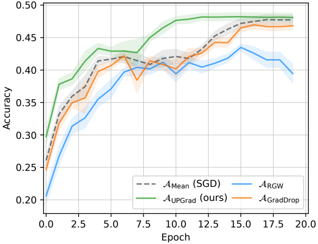

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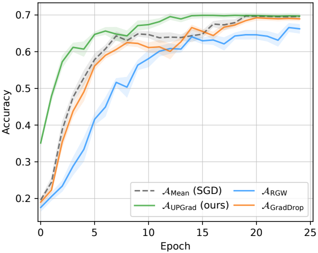

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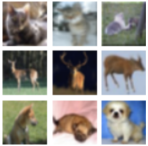

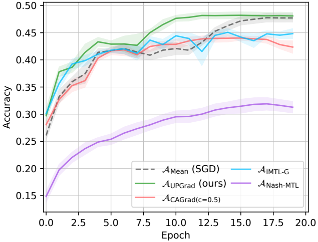

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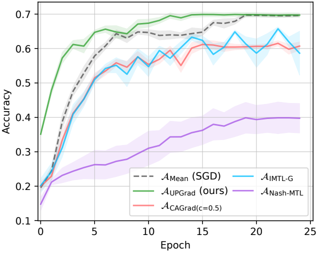
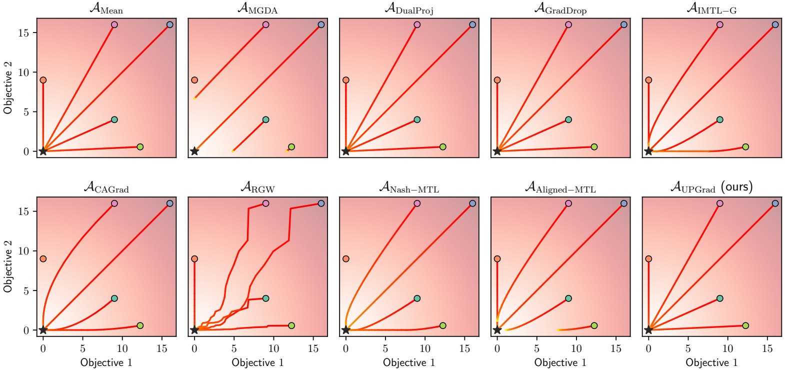


Element-wise quadratic function trajectories

Minimize $\boldsymbol{f} : \begin{bmatrix} x \\ y \end{bmatrix} \mapsto \begin{bmatrix} x^2 \\ y^2 \end{bmatrix}$. It's Jacobian at $\begin{bmatrix} x \\ y \end{bmatrix}$ is $\begin{bmatrix} 2x & 0 \\ 0 & 2y \end{bmatrix}$.



Convex quadratic form trajectories

Some convex quadratic function $\boldsymbol{f} : \mathbb{R}^2 \to \mathbb{R}^2$, with conflicting gradients.